A

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CERTIFICATE

This is to certify that the Project Phase-II Report entitled "**Design and Performance Evaluation of an Intelligent Adaptive Digital Communication Receiver using Automatic Modulation Classification**" is presented & submitted by Hari Gupta, Virav Vanani, Surendra Gamit, bearing Roll No. U23EC166, U23EC182, U23EC195 of B.Tech. Final Year in the partial fulfillment of the requirement for the award of B.Tech. Degree in Electronics & Communication Engineering for academic year 2026-27.

They have successfully and satisfactorily completed their **Project Phase-II Evaluation** in all respects. We certify that the work is comprehensive, complete and fit for evaluation.

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Declaration

We hereby declare that the work presented in this report is original and has been carried out by us under the guidance of **Dr. Golak Santra**. The design, implementation, experiments, analysis, and documentation presented here are prepared for the Phase-II evaluation of the B.Tech. project. This report has not been submitted, either in part or in full, to any other university or institution for the award of any degree or diploma.

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Abstract

Automatic modulation classification is useful in receivers that must operate without complete prior knowledge of the transmitted waveform. The objective of this Phase-II project is to extend modulation recognition into a complete adaptive digital communication receiver. Instead of ending at a modulation label, the proposed system uses the estimated format to configure a demodulator and evaluates whether the transmitted information is recovered correctly.

The transmitter generates random information bits and maps them to BPSK, QPSK, 8-PSK, or 16-QAM symbols. The symbols pass through a block-Rayleigh fading and additive-white-Gaussian-noise channel with carrier phase and frequency offsets. At the receiver, pilot-assisted synchronization and frame-level channel equalization are applied. A penalized nearest-constellation estimator selects the modulation, after which the corresponding demodulator recovers the bits. The experiment records bit error rate, symbol error rate, modulation-decision accuracy, packet success rate, and effective throughput over an SNR range of -4 to 16 dB.

The final experiment uses 200 frames per SNR and a fixed seed of 42. At 16 dB, the receiver achieves 0.095% BER, 0.172% SER, 100% modulation-decision accuracy, 75.5% packet success, and 1.888 bits/symbol effective throughput. The results show that an AMC decision must be evaluated through the complete information path because a correct format decision and error-free bit recovery are different requirements. The report documents the Phase-II theory, communication architecture, algorithms, experimental setup, measured results, limitations, and future SDR extensions.

List of Abbreviations

AMC	Automatic Modulation Classification
AWGN	Additive White Gaussian Noise
BER	Bit Error Rate
CSI	Channel State Information
I/Q	In-phase and Quadrature
PHY	Physical Layer
QAM	Quadrature Amplitude Modulation
PSK	Phase Shift Keying
RF	Radio Frequency
SDR	Software Defined Radio
SER	Symbol Error Rate
SNR	Signal-to-Noise Ratio

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Chapter 1
Introduction

1.1 Motivation and Background

Modern wireless systems operate in congested and dynamic spectrum environments. The transmitter may change its modulation, the channel may introduce fading and phase rotation, and several users may share adjacent frequency bands. A conventional receiver is normally configured with the modulation and synchronization parameters before data reception begins. This assumption is unsuitable for a non-cooperative receiver, a cognitive-radio node, or a monitoring station that must process unknown signals.

Automatic modulation classification is the task of identifying the modulation format from received samples. It supports adaptive demodulation, spectrum monitoring, electronic support systems, interference analysis, and waveform-aware resource allocation. Classical approaches use likelihood tests, higher-order moments, cumulants, and cyclostationary features. These methods are physically interpretable, but their performance depends on the observation length, channel assumptions, and quality of synchronization [1, 2, 4].

Deep learning has provided another route by learning signal representations directly from I/Q samples. A neural classifier can identify local temporal patterns and phase relationships without requiring a separate handcrafted feature-design stage [7, 11]. However, a classification accuracy value alone does not demonstrate that a receiver can recover the transmitted information. The communication consequence of a wrong modulation decision must be measured after demodulation.

1.2 Applications of AMC

AMC is relevant to cognitive radio, where a receiver senses a waveform and selects a compatible processing chain. It is also useful in spectrum surveillance, where several unknown signals must be characterized from short observations. In adaptive wireless systems, modulation recognition can support mode selection, link adaptation, and interference-aware processing. In software-defined radio, the classifier can act as a control block between the digitized signal and a bank of candidate demodulators.

These applications impose requirements that are different from a static classification benchmark. The receiver must operate across channel conditions, report

confidence or uncertainty, and provide a measurable benefit to the recovered information. Phase II is designed around this system-level requirement.

1.3 Receiver Requirements

The receiver developed in this phase is required to perform more than a single classification operation. It must accept a stream of complex samples, estimate the modulation format, select a compatible demodulator, and report the quality of the recovered information. This requirement changes the way the experiment is designed. The transmitted bits must remain available as a reference, channel effects must be represented explicitly, and the output must include communication metrics rather than only a class label.

The receiver should also behave consistently when the signal quality changes. At high SNR, it should identify the modulation and recover most of the information bits. At low SNR, it should expose its uncertainty instead of producing a misleadingly confident output. For this reason, decision accuracy, BER, SER, packet success, and throughput are recorded separately. Together they show whether a recognition decision is useful to the physical layer.

1.4 Problem Definition and Objectives

Problem Statement: Design and evaluate an adaptive digital communication receiver that estimates the transmitted modulation and uses that estimate to recover information bits under noise, fading, phase offset, and frequency offset.

The objectives of the project are:

1. Generate reproducible information frames using BPSK, QPSK, 8-PSK, and 16-QAM.
2. Model a block-Rayleigh and AWGN channel with carrier impairments.
3. Apply pilot-assisted synchronization and channel equalization.
4. Estimate the modulation format using received constellation structure.
5. Select a compatible demodulator automatically.
6. Measure BER, SER, packet success, throughput, and decision accuracy.
7. Compare adaptive processing with a fixed-QPSK receiver.
8. Document the Phase-II implementation and experimental results clearly.

1.5 Contributions

The main contribution is a complete transmitter–channel–receiver simulation in which modulation recognition controls demodulation. The work adds four digital modulation formats, configurable channel impairments, frame-level correction, an interpretable estimator, adaptive bit recovery, and a fixed-modulation reference. It also introduces communication-level evaluation rather than reporting only a class label.

1.6 Scope and Limitations

The current work is a controlled software simulation. It includes a flat block-fading model, AWGN, carrier phase and frequency offsets, and pilot-assisted correction. Pulse shaping, matched filtering, timing-loop implementation, forward-error correction, interleaving, retransmission, and over-the-air SDR measurements are outside the present scope. These limitations are stated so that the simulation results are not confused with hardware measurements.

1.7 Organization of the Report

Chapter 2 reviews classical AMC, deep-learning approaches, communication receiver design, and the research gap. Chapter 3 presents the mathematical background of complex baseband signals, digital modulation, channel impairments, synchronization, and evaluation metrics. Chapter 4 describes the generated Phase-II data, modulation classes, preprocessing, and experimental record. Chapter 5 explains the proposed receiver methodology and design. Chapter 6 describes implementation and experimental setup. Chapter 7 presents measured results and discussion. Chapter 8 concludes the work and identifies future extensions.

1.8 Chapter Summary

Chapter 1 establishes the motivation, project objectives, and distinction Chapter 2 places the work in the context of classical AMC, learning-based recognition, and adaptive receiver design. Chapter 3 develops the signal, channel, synchronization, and metric equations. Chapter 4 describes how the communication frames are generated and prepared. Chapter 5 gives the proposed receiver in block form. Chapter 6 records the software and experiment settings. Chapter 7 presents the

complete Phase-II SNR sweep and receiver comparison. Chapter 8 discusses the contribution, limitations, and next implementation steps.

Chapter 2
Literature Review

2.1 Classical AMC Approaches

Early AMC systems used statistical moments, higher-order cumulants, likelihood functions, and cyclostationary features. Higher-order statistics can distinguish modulation families because the distribution of a received signal changes with the waveform. Cyclostationary methods exploit periodic features introduced by symbol timing, carrier structure, or framing [4, 5].

The advantage of classical methods is interpretability. A designer can identify which feature is responsible for a decision and can often estimate the computational complexity analytically. Their limitation is sensitivity to observation length and model mismatch. Noise, fading, carrier offset, frequency-selective propagation, and nonlinear hardware can alter the statistics on which the decision depends.

2.2 Likelihood and Feature-Based Methods

Likelihood-based AMC compares the received sequence with a probability model for every candidate modulation. When the channel and noise parameters are known, such methods can be statistically efficient. In practical systems, the required likelihood model is difficult to maintain because the receiver may not know the exact fading, synchronization, or hardware condition.

Feature-based systems reduce the computational burden by extracting moments, spectral features, cyclic features, or instantaneous amplitude and phase statistics. They can provide a useful compromise between complexity and performance, but their success depends on choosing features that remain discriminative at low SNR. Phase II uses a constellation-based estimator because it is transparent and directly connected to demodulator selection.

2.3 Deep Learning for AMC

Deep learning changed the AMC workflow by allowing features to be learned from raw I/Q sequences. Convolutional neural networks are attractive because one-dimensional filters can detect local waveform patterns, short phase transitions, and amplitude relationships. RadioML-style datasets have provided a common basis for studying these models [7, 8].

The main limitation of a classifier-only evaluation is that the output is often treated as the final result. A high classification accuracy does not directly specify the BER after a receiver uses the decision. Conversely, an incorrect decision may select an incompatible demodulator and cause a much larger information loss than a single classification error suggests.

2.4 Adaptive Receiver Design

An adaptive receiver contains a sensing or recognition stage followed by a configuration stage. The configuration stage may select a demodulator, symbol mapping, coding mode, timing method, or channel-estimation strategy. This makes receiver performance dependent on both recognition and subsequent signal processing.

The Phase-II design follows this principle. The estimator does not operate as an isolated classifier. Its output selects one of four demodulators, and the receiver compares the recovered bits with the transmitted bits. This creates a traceable relationship between waveform uncertainty and communication reliability.

2.5 Over-the-Air Generalization

Synthetic experiments provide repeatability and controlled comparison, but they do not include every effect present in an RF front-end. Oscillator phase noise, automatic-gain-control transients, antenna response, analog filtering, frequency-selective multipath, and timing drift can reduce performance when a system moves to an SDR. Over-the-air studies have shown the importance of these effects [8].

The present report therefore treats the simulation as an engineering validation stage. The channel model is sufficiently rich to expose modulation-decision and bit-recovery failures, while remaining reproducible for a project evaluation. Hardware validation is proposed as the next stage.

2.6 Benchmark Datasets and Tooling

RadioML 2016.10a is a widely used benchmark for raw-I/Q modulation classification. A classification-only benchmark does not provide transmitted-bit references, so the Phase-II link is evaluated with generated information frames so that BER and packet success can be measured directly. Python, NumPy, and Matplotlib provide the numerical and visualization environment used by the project.

2.7 Summary of Representative Studies

Table 2.1: Representative studies and their relation to the Phase-II work.

Reference	**Approach**	**Relevance to this project**
Gardner [5]	Cyclostationary analysis	Provides interpretable signal features for detection.
Azzouz and Nandi [4]	Statistical AMC	Shows the importance of feature and channel assumptions.
O'Shea and Corgan [7]	CNN on raw I/Q	Motivates data-driven waveform recognition.
O'Shea *et al.* [8]	Over-the-air classification	Demonstrates the need for realistic validation.
This project	Adaptive communication receiver	Connects modulation estimation to demodulation and BER.

2.8 Research Gap

Many AMC studies end after predicting a modulation label, while many link simulations assume that the modulation is already known. The engineering gap between these two stages is whether the recognition result can configure a useful receiver and deliver reliable information. Phase II addresses this gap by evaluating the complete path from generated bits to recovered bits.

2.9 Comparison of Evaluation Practices

Three evaluation practices can be distinguished in the reported literature. The first reports only a classifier accuracy, which is useful for comparing recognition models but does not show how the decision affects a receiver. The second assumes perfect modulation knowledge and evaluates BER, which gives a clear link result but removes the recognition problem. The third combines recognition with demodulation and evaluates the recovered information. The present project follows the third practice.

This distinction is important when comparing numerical values. A classifier accuracy near one does not imply a BER near zero because symbol decisions still depend on residual noise, gain error, phase error, and the separation between

constellation points. Similarly, a low BER averaged over all symbols may still produce a low packet-success rate when a long frame is declared unsuccessful after one error. The report therefore avoids using one metric as a substitute for the complete receiver result.

2.10 Design Choices from the Review

The review led to four design choices. First, the modulation set contains both phase-only formats and a rectangular amplitude-and-phase format. This allows the receiver to be tested on different decision geometries. Second, the channel contains both additive noise and multiplicative fading so that the estimator is not tested only on an ideal amplitude scale. Third, a fixed-QPSK path is included as a simple reference rather than as a competing learning model. Finally, the estimator is based on constellation distance, which makes each decision traceable and leaves a clear path for replacing it with a trained model in future work.

2.11 Summary

The literature supports the need for automatic modulation recognition, but it also shows that a classification result is only one part of a receiver. The Phase-II work adds a controlled communication link around modulation recognition. This arrangement makes the effect of modulation estimation visible in the final bit and packet metrics.

Chapter 3
Theoretical Background

3.1 Baseband Signal Model

A passband signal can be represented in complex baseband form as

$$x[n] = I[n] + jQ[n], \tag{3.1}$$

where $I[n]$ and $Q[n]$ are the in-phase and quadrature components. The complex representation preserves amplitude and phase information while allowing the transmitter, channel, and receiver to be implemented as discrete-time operations.

For a frame of N symbols, the transmitter generates $x[0], x[1], \ldots, x[N-1]$. The receiver observes a corrupted sequence and must estimate the format before applying a symbol-to-bit mapping. The quality of this estimate affects every subsequent stage.

3.2 Digital Modulation Families

BPSK uses two opposite phase points and carries one bit per symbol. Its large decision separation makes it robust, but its spectral efficiency is low. QPSK uses four phase points and carries two bits per symbol, providing a common balance between robustness and rate. 8-PSK carries three bits per symbol by reducing angular separation. 16-QAM uses both amplitude and phase and carries four bits per symbol, but requires more accurate gain, phase, and noise control.

Table 3.1: Digital modulation formats used by the Phase-II receiver.

Format	**Order**	**Bits/symbol**	**Characteristic**
BPSK	2	1	Two phase states and large decision margin.
QPSK	4	2	Four phase states and moderate spectral efficiency.
8-PSK	8	3	Higher rate with reduced angular separation.
16-QAM	16	4	Amplitude and phase states for high rate.

3.3 Channel Impairments

The Phase-II received signal is modelled as

$$y[n] = h\, x[n] e^{j(2\pi \Delta f n + \phi)} + w[n], \tag{3.2}$$

where h is a complex block-Rayleigh channel coefficient, Δf is a normalized frequency offset, ϕ is a carrier phase offset, and $w[n]$ is complex AWGN. The coefficient is constant during one frame and changes between frames, allowing fading to affect packet-level performance.

Rayleigh fading changes the amplitude and phase of the signal without a dominant line-of-sight component. AWGN introduces random perturbations whose variance is controlled by the selected SNR. Phase offset rotates the entire constellation, while frequency offset produces a rotation that changes with sample index. These impairments test different aspects of receiver correction and constellation recognition.

3.4 Effect of Individual Channel Terms

The four channel terms affect the receiver in different ways. AWGN moves each sample independently and changes the spread of the constellation. Block-Rayleigh fading changes the complex gain of the complete frame and can make one frame substantially weaker than another at the same nominal SNR. A phase offset produces a common rotation, while a frequency offset produces a rotation that changes with sample index.

These effects are not interchangeable. A gain error mainly changes the distance from the origin, whereas a carrier error changes the angular position of the points. A receiver that corrects only amplitude can still make a poor decision when the constellation is rotated. Likewise, a carrier correction without channel equalization can leave the amplitudes of 16-QAM points incorrectly scaled. The Phase-II sequence therefore corrects both the carrier terms and the block channel before estimating the format.

3.5 Equalization Error and Noise Enhancement

The equalized sample is obtained by dividing by the estimated channel coefficient. When the magnitude of the coefficient is small, this division also amplifies the noise component. This is a known trade-off of zero-forcing equalization. A practical receiver could use a regularized equalizer or a confidence flag when the channel

estimate is too weak.

The present experiment keeps the equalization rule fixed across all SNR values. This makes the result useful for studying the adaptive decision rather than hiding a weak channel realization through a changing equalization parameter. The resulting high-SNR error floor is discussed in Chapter 7 as a consequence of the complete receiver chain.

3.6 Why a Common Power Reference is Required

The modulation formats carry different numbers of bits per symbol and have different geometries. If their average powers were not normalized, a format with a larger amplitude would receive an artificial SNR advantage. Power normalization makes the comparison focus on point separation, channel distortion, and receiver decisions. It also makes the throughput comparison meaningful because a larger rate is not obtained simply by transmitting a larger signal.

3.7 Signal-to-Noise Ratio

The signal-to-noise ratio is defined as

$$\mathrm{SNR}_{dB} = 10 \log_{10} \left(\frac{P_s}{P_n} \right), \tag{3.3}$$

where P_s is the average signal power and P_n is the average noise power. The simulation normalizes the candidate constellations and scales complex Gaussian noise from the selected SNR so that the modulation formats are compared under one common convention.

3.8 Synchronization and Equalization

In a practical receiver, synchronization includes automatic gain control, timing recovery, carrier-frequency estimation, carrier-phase recovery, and channel estimation. Phase II uses a pilot-assisted simulation estimate for the frame-level carrier and channel parameters. This isolates the new adaptive-modulation problem from an unfinished hardware synchronization loop.

After carrier correction, the frame is equalized using

$$\hat{x}[n] = \frac{y[n]}{h}. \tag{3.4}$$

The corrected samples are normalized before their distance to each candidate constellation is calculated. The limitation of this controlled estimate is explicitly discussed in the results chapter.

3.9 AMC Approaches

The project distinguishes three receiver actions. A fixed receiver applies one known demodulator to every frame. A classifier-only receiver produces a label but does not recover information. The proposed adaptive receiver estimates the format and passes the decision to the corresponding demodulator. This third arrangement is the focus of Phase II.

3.10 Communication Metrics

The bit error rate and symbol error rate are

$$\mathrm{BER} = \frac{N_{\mathrm{bit,error}}}{N_{\mathrm{bit,total}}}, \qquad \mathrm{SER} = \frac{N_{\mathrm{symbol,error}}}{N_{\mathrm{symbol,total}}}. \tag{3.5}$$

A packet is successful only when every bit in its frame is recovered correctly. If b_m is the bits-per-symbol value of the selected modulation, the effective throughput is the average of b_m multiplied by the packet-success indicator. Therefore, packet success is stricter than BER and exposes the practical effect of even a small number of bit errors.

3.11 Bit Mapping and Symbol Decisions

For a modulation order M, the input stream is divided into groups of $\log_2 M$ bits. Each group is assigned to one point in the normalized constellation. The same mapping is used in reverse at the receiver after a nearest-point decision. Using one mapping definition in both directions avoids an artificial error caused by a mismatch between the transmitter and demodulator.

The minimum distance between constellation points is a useful qualitative indicator of robustness. BPSK has the largest separation for its rate, while 8-PSK packs more points around the same phase circle. 16-QAM provides four bits per symbol, but its inner points can be confused with adjacent points after noise and amplitude distortion. These differences explain why one common SNR does not produce the same BER for all formats.

3.12 Frame-Level Reliability

If the probability of a bit error is p_b and a frame contains L bits, the idealized probability of a completely correct frame is approximately

$$P_{\text{packet}} \approx (1 - p_b)^L. \tag{3.6}$$

The approximation assumes independent bit errors and is not used as a replacement for the measured packet count. It explains, however, why a BER of a few percent can coexist with a packet-success rate near zero for a long frame. In the experiment, the frame length changes with modulation order because the symbol count is fixed, so the measured packet metric also reflects the information load of each format.

3.13 Throughput Definition

The nominal information rate of a modulation is its bits-per-symbol value. The effective value reported in this project multiplies that value by the packet-success indicator before averaging over frames:

$$T_{\text{eff}} = \frac{1}{F} \sum_{i=1}^{F} b_i \mathbb{1}\{\text{frame } i \text{ is correct}\}. \tag{3.7}$$

This definition does not include coding overhead, pilot overhead, filtering loss, or hardware latency. It is intended to compare the useful information delivered by the adaptive and fixed receivers under the same simulation conditions.

3.14 Interpretation of Decision Accuracy

Decision accuracy measures whether the estimated modulation label matches the transmitted label. It does not measure whether all symbols are correct. For example, at 10 dB the measured decision accuracy is 99.5 percent, while the BER is 4.69 percent. This difference is expected because the estimator can identify the family correctly while individual noisy samples cross decision boundaries. The separation of these two measurements is one of the main reasons for using a complete link experiment.

Chapter 4
Dataset and Preprocessing

4.1 Phase-II Data Generation

Unlike a classification-only experiment, the Phase-II link begins with information bits rather than pre-labelled signal windows. For every frame, a reproducible random generator creates a binary sequence. One of the four candidate modulation formats is selected, and the bits are grouped according to its order. The resulting complex symbols are retained as the transmitter reference for later error calculation.

Each frame contains 256 symbols. Thus, BPSK frames contain 256 information bits, QPSK frames contain 512 bits, 8-PSK frames contain 768 bits, and 16-QAM frames contain 1024 bits. Using the same symbol-frame length makes packet-level comparisons consistent while allowing the information rate to change with modulation order.

4.2 Modulation Classes

Table 4.1: Modulation classes used to generate Phase-II communication frames.

Class	**Bits/symbol**	**Use in the experiment**
BPSK	1	Robust low-order reference.
QPSK	2	Fixed-receiver comparison format.
8-PSK	3	Higher-order phase modulation.
16-QAM	4	High spectral-efficiency format.

4.3 Constellation Normalization

The constellation points are normalized to a common average-power convention. This avoids giving one format an artificial advantage because of a larger absolute amplitude. After channel correction, the received symbols are placed on the same scale before nearest-point distances are compared.

4.4 Channel Data Preparation

For each generated frame, the channel draws a block-Rayleigh coefficient, carrier phase, frequency offset, and complex AWGN realization. The SNR is selected from the sweep

$$-4, -2, 0, 2, 4, 6, 8, 10, 12, 14, 16 \text{ dB}.$$

The transmitted bits, transmitted symbols, received samples, estimated format, recovered bits, and error counts are retained until all frame-level metrics have been calculated.

4.5 Preprocessing at the Receiver

The receiver applies the inverse simulated carrier rotation and equalizes the block channel before estimation. The corrected samples are normalized using the candidate constellation scale. This operation is not a replacement for a practical timing loop; it is a controlled preprocessing stage that makes the contribution of adaptive modulation measurable.

4.6 Experiment Split and Repetition

The communication experiment is not divided into train, validation, and test subsets because it is a link simulation rather than a supervised training run. Instead, every SNR value is evaluated using 200 independently generated frames with seed 42. The same modulation list, frame length, channel model, and metric definitions are used at every SNR.

4.7 Data Integrity and Reproducibility

The complete numerical arrays are written to a JSON record. This record contains the SNR values, adaptive and fixed-receiver BER and SER, decision accuracy, packet success, throughput, modulation list, frame count, and seed. Figures are generated from the same record, preventing manual transcription errors between the experiment and the report.

4.8 Limitations of the Generated Data

The data is generated by a controlled software channel and therefore does not include antenna response, analog filtering, oscillator phase noise, frequency-selective

multipath, timing drift, or hardware nonlinearity. The generated frames are appropriate for algorithm verification and comparative analysis, but a future SDR experiment is required before making claims about over-the-air performance.

4.9 Data Record Structure

Every frame has a corresponding record containing the selected modulation, SNR, channel realization, transmitted bits, received samples, estimated modulation, recovered bits, and error counts. The experiment driver reduces these frame records into arrays indexed by SNR. Keeping the frame-level information during the experiment is useful for diagnosing unusual results, such as a high decision accuracy together with a low packet-success rate.

The final JSON file stores the aggregate values used in the report. It also stores the modulation names and the experiment seed, so the order of the curves is not dependent on the order in which a plotting script happens to read a dictionary. The PNG figures are presentation views of this record; they are not manually redrawn result diagrams.

4.10 Normalization Checks

Before the channel is applied, the average power of each normalized constellation is checked. This check is important because the SNR sweep is meaningful only when the formats are compared at a common reference power. After equalization, the received samples are normalized using the same convention before distance scores are calculated. A gain mismatch would otherwise make the estimator prefer a format for amplitude reasons rather than for its geometric structure.

4.11 Quality-Control Cases

Three simple cases are used before the full sweep. In the first case, a clean frame is modulated and demodulated without a channel; all recovered bits must match. In the second case, an invalid modulation name or an incompatible bit length must produce an explicit validation error. In the third case, a clean constellation is supplied to the estimator; the expected class must be selected. These cases do not replace the noisy experiment, but they separate implementation errors from channel-induced errors.

Chapter 5
Proposed Methodology and Model Design

5.1 End-to-End Pipeline

The proposed Phase-II pipeline is

$$\begin{aligned}\text{random bits} \rightarrow \text{modulator} \rightarrow \text{channel} \rightarrow \text{correction} \rightarrow \text{AMC decision} \rightarrow \\ \text{adaptive demodulator} \rightarrow \text{recovered bits.}\end{aligned} \tag{5.1}$$

The final bit sequence is compared with the transmitter reference. This pipeline is intentionally different from a classifier-only workflow because the modulation decision controls a subsequent physical-layer operation.

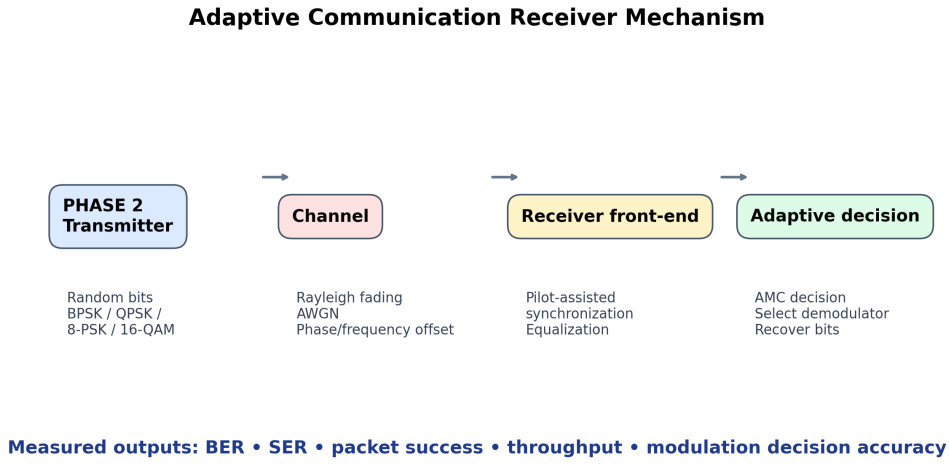


Figure 5.1: End-to-end Phase-II adaptive communication receiver.

5.2 Transmitter Model

The transmitter selects a modulation and groups the random bits into symbols. The mapper produces normalized complex points according to the selected constellation. The mapping is deterministic and is reused by the corresponding demodulator, so a clean round-trip test should recover all bits exactly.

5.3 Channel Model

The channel applies block fading, carrier rotation, and AWGN in sequence. Block fading changes the amplitude and phase of a complete frame. Carrier phase offset rotates the constellation, while frequency offset causes a progressive rotation. AWGN controls the SNR-dependent uncertainty that eventually produces both estimator errors and symbol errors.

5.4 Receiver Front-End

The receiver front-end reverses the simulated carrier rotation and divides by the block channel coefficient. The output is a corrected complex sequence. In an SDR implementation, this stage would be replaced by a timing recovery loop, pilot detector, carrier-recovery loop, and channel estimator. The current design keeps the estimate controlled so that the adaptive demodulation contribution can be isolated.

5.5 Constellation-Distance Estimator

Let $\mathcal{C}_m$ denote the normalized constellation for candidate format m. For each corrected received sample, the estimator finds the nearest point in $\mathcal{C}_m$ and calculates the average distance d_m. It then selects

$$\hat{m} = \arg\min_{m} \left(d_m + \lambda \log_2 |\mathcal{C}_m| \right), \tag{5.2}$$

where the penalty discourages a high-order constellation from winning only because it contains more possible points.

5.6 Adaptive Demodulator

The estimated format selects one of four demodulation rules. Each corrected sample is assigned to its nearest point, and the point index is converted back to bits using the transmitter's mapping. The recovered bits are aligned with the original frame and compared bit by bit. The estimator decision is recorded separately so that decision errors and within-format detection errors can be distinguished.

5.7 Fixed-QPSK Baseline

The fixed-QPSK receiver applies the QPSK demodulator regardless of the transmitted format. It represents a system that does not perform modulation adaptation. Its performance is expected to remain poor for BPSK, 8-PSK, and 16-QAM frames, but it provides a useful reference for measuring the value of adaptive selection.

5.8 Failure Propagation

The system has a clear failure chain. At low SNR, constellation points overlap and the estimator may choose the wrong format. The wrong format selects an incompatible demodulator, which can create many bit errors. Even when the estimated format is correct, noise can move individual points across decision boundaries. A single bit error is sufficient to mark a complete packet unsuccessful.

5.9 Design Rationale

The Phase-II estimator is deliberately interpretable. The candidate constellations, distance scores, selected modulation, recovered symbols, and final bit metrics can all be inspected. This design also provides a baseline for future replacement by a trained CNN or a confidence-aware estimator.

5.10 Detailed Receiver Sequence

The receiver processes one frame in a fixed order. It first removes the known simulated carrier rotation and compensates the block channel. It then computes candidate distances for all four constellations. The minimum penalized distance determines the modulation estimate. Only after this decision is made does the receiver select the corresponding bit mapper. Finally, the recovered bit sequence is compared with the transmitter reference and the frame-level counters are updated.

This sequence prevents the experiment from accidentally using the true transmitted modulation during demodulation. The true label is used only for measuring decision accuracy and for calculating the reference errors. The demodulator receives the estimate produced by the receiver, which makes the adaptive path a genuine decision-dependent pipeline.

5.11 Estimator Penalty and Bias

The number of points in a constellation affects the average nearest-point distance. A high-order format has more candidate points and can sometimes obtain a lower unpenalized distance even when the received samples are better explained by a lower-order format. The penalty term in the estimator reduces this bias by adding a cost proportional to the logarithm of the constellation order.

The penalty is not intended to be a universal statistical criterion. It is a simple engineering correction that keeps the estimator interpretable. Its value is held constant over the SNR sweep so that the improvement at high SNR is not produced by changing a tuning parameter at each point. A future version can select this parameter using a separate validation experiment or replace the distance rule with a learned confidence model.

5.12 Error Propagation Paths

There are two main paths from a noisy sample to an incorrect frame. In the first path, the constellation estimator selects the wrong format. The demodulator then uses an incompatible geometry and a large number of bits can be wrong even when individual samples are not extremely noisy. In the second path, the format estimate is correct but one or more samples cross a decision boundary. This produces symbol and bit errors without reducing decision accuracy.

5.13 Complexity Considerations

For a frame containing N received symbols and candidate constellations $\mathcal{C}_m$, the distance calculation requires approximately

$$\mathcal{O}\left(N \sum_m |\mathcal{C}_m|\right)$$

complex comparisons. With the four formats used here, the largest term is the 16-QAM candidate set. The implementation is therefore suitable for offline experiments and small real-time demonstrations. A deployed receiver could reduce the cost using a hierarchical classifier, moment-based pre-screening, or a neural model followed by a small verification step.

5.14 Algorithmic Procedure

The receiver applies the same sequence to every frame, independent of the transmitted format. First, the received samples are corrected using the simulated frame-level estimates. Next, the distance score is calculated for each candidate constellation. The candidate with the smallest penalized score is selected, and only then is the corresponding demodulator called. The recovered bits are aligned with the reference bits before the error counters are updated.

This order is important for a fair adaptive experiment. If the true modulation label were passed directly to the demodulator, the experiment would measure a known-modulation receiver rather than an adaptive one. The true label is used only after recovery, for decision-accuracy calculation and for grouping the results during analysis.

5.15 Confidence and Decision Margin

The difference between the smallest and second-smallest candidate scores can be used as a simple decision margin. A large margin indicates that one constellation explains the corrected samples better than the alternatives. A small margin indicates ambiguity, even if the estimator must still return one modulation for the present experiment.

The current result record does not use a confidence threshold to reject frames, but the margin provides a direct extension for future work. A receiver could declare an uncertain frame, select a robust fallback demodulator, or request additional observations. Such a confidence layer would be especially useful in the 8–10 dB transition region identified in Chapter 7.

5.16 Channel and Receiver Consistency

The simulated correction stage uses the same channel parameter convention as the channel generator. This makes it possible to isolate modulation selection and bit recovery. In a practical receiver, the correction values would be estimates and would contain error. The separation is still useful because it establishes a reference experiment before timing-loop and channel-estimation imperfections are added.

Chapter 6
Implementation and Experimental Setup

6.1 Software and Tools

The project is implemented in Python using NumPy for numerical operations, Matplotlib for figures, and pytest for automated verification. The source is organized into reusable communication primitives, an experiment driver, visualization utilities, tests, and report artifacts. The report does not include executable command blocks; the repository documentation contains the operational instructions.

6.2 Experimental Environment

The final communication experiment is deterministic at the experiment level. The random seed is 42, the frame length is 256 symbols, and 200 frames are processed at each SNR. The channel and estimator parameters are held constant across the sweep, allowing changes in the curves to be interpreted primarily as SNR effects.

6.3 Hyperparameters and Link Parameters

6.4 Evaluation Metrics

Five primary metrics are calculated. BER counts erroneous bits, SER counts erroneous symbols, and modulation-decision accuracy counts correct format estimates. Packet success requires every bit in a frame to be correct. Effective throughput weights the bits-per-symbol value of each successful frame by its packet-success indicator. These metrics are reported together because no single value captures the complete receiver behavior.

6.5 Experimental Procedure

At each SNR, the experiment generates 200 frames. For every frame it selects a modulation, creates bits, maps them to symbols, applies the channel, corrects the received sequence, estimates the modulation, demodulates the symbols, and calculates errors. The same transmitted frame is also processed by the fixed-QPSK reference. Aggregate arrays are saved before figures are created.

Table 6.1: Phase-II link parameters used for the final experiment.

Parameter	**Value**
Modulation formats	BPSK, QPSK, 8-PSK, 16-QAM
Symbols per frame	256
SNR range	-4 to 16 dB
SNR increment	2 dB
Frames per SNR	200
Random seed	42
Channel	Block Rayleigh fading and AWGN
Carrier impairments	Phase and frequency offset
Correction	Pilot-assisted simulation estimate
Estimator	Penalized nearest-constellation distance

6.6 Verification

Automated tests cover clean modulation round trips, validation of invalid inputs, deterministic data handling, and clean constellation identification. The report source is checked for balanced LaTeX environments, existing figure references, and the required style dependencies. These checks supplement the numerical experiment.

6.7 Reproducibility Record

The JSON result record stores the seed, frame count, SNR list, modulation list, adaptive metrics, and fixed-QPSK metrics. The figures are generated from the same values used in the tables.

6.8 Implementation Modules

The implementation is divided into modulation, channel, receiver, metric, and experiment-driver responsibilities. The modulation functions define the constellations and bit mappings. The channel functions apply fading, carrier offset, and noise. The receiver functions perform correction, estimate the format, and demodulate. The experiment driver repeats these operations over the SNR list and

writes the aggregate record.

This separation is useful during debugging because each stage can be tested without running the complete sweep. It also makes the Phase-II design different from a single notebook-style experiment. The report presents the algorithmic behaviour without reproducing source-code listings, while the repository retains the implementation files used to produce the numerical results.

6.9 Numerical Conventions

Complex samples use NumPy complex arrays and the real and imaginary parts represent the in-phase and quadrature components. Random numbers are generated from a fixed seed at the experiment level. SNR values are stored in decibels, while noise scales are calculated in linear power units before being applied to the complex samples. Percentages in the report are rounded for readability; the JSON record retains the underlying decimal values.

6.10 Parameter Selection

The frame length of 256 symbols gives enough samples for a constellation estimate while keeping one experiment run manageable. Two hundred frames per SNR provide a meaningful packet-success count without making the experiment dependent on a single channel realization. The SNR interval from -4 to 16 dB covers the transition from an unusable low-SNR link to a high-SNR link where modulation decisions are stable.

6.11 Setup Limitations

Repeating the experiment with the same seed reproduces the stored numerical record. Repeating it with a different seed is expected to change individual frames and slightly change the aggregate values, especially packet success, because packet success is a binary frame-level metric. The experiment is therefore deterministic for comparison, but the result should not be read as an exact performance guarantee for every random channel realization.

6.12 Runtime and Resource Considerations

The communication experiment is dominated by the repeated distance calculation for the candidate constellations. Its memory requirement is small because

one frame is processed before the aggregate arrays are updated. This makes the experiment suitable for an ordinary project computer and avoids making the reported result dependent on a specialised GPU. The recognition stage and the Phase-II communication link are kept as separate modules with separate resource requirements.

The same implementation can be extended in two directions. More frames can be evaluated to improve statistical resolution, or more candidate formats and channel models can be added to test generality. In both cases the result record should be extended before the report tables are updated so that every new curve remains traceable to an experiment setting.

6.13 Frame Processing and Storage

One frame is processed completely before the aggregate arrays are updated. The transmitter bits and symbols are kept as the reference, while the received and corrected arrays are temporary processing values. The experiment stores aggregate metrics by SNR rather than storing every noisy sample in the final result file. This keeps the numerical record compact while preserving the values needed for all report figures and tables.

6.14 Input and Output Checks

Before the link experiment begins, the implementation checks the modulation name, the number of bits required for the selected symbol mapping, the frame length, and the dimensions of the complex arrays. After processing, the number of recovered bits is checked against the number of transmitted bits. These checks prevent a shape error from appearing as a communication result.

6.15 Experiment Repetition Strategy

The SNR sweep is repeated with the same frame count and modulation selection rule at every point. The channel parameters are redrawn for each frame, while the experiment seed fixes the overall sequence of draws. This gives each SNR point the same number of opportunities for a successful packet and makes the change in the curves primarily a result of channel quality.

6.16 Implementation Limitations

The present code is intentionally focused on the communication path. It does not yet implement a real-time stream buffer, timing recovery from an unknown sample clock, or a hardware interface. It also uses one frame-level fading coefficient rather than a time-varying multipath model. These are implementation boundaries for the current phase and form direct requirements for the next receiver version.

6.17 Software Module Responsibilities

Table 6.2: Responsibilities of the main Phase-II software modules.

Module	**Responsibility**
Constellation mapping	Converts bit groups to normalized complex symbols.
Channel model	Applies fading, offsets, and complex AWGN.
Correction stage	Removes the simulated carrier and channel effects.
AMC estimator	Scores candidate constellations and selects one format.
Demodulator	Maps corrected symbols back to bit groups.
Metric calculation	Computes BER, SER, packet success, and throughput.
Experiment driver	Repeats frames over the SNR sweep and writes results.

Keeping these responsibilities separate allows a future timing loop or learned estimator to replace one stage without changing the metric definitions. It also makes it possible to test a clean mapper and demodulator before the channel is introduced.

Chapter 7
Results and Discussion

7.1 Training Dynamics

Phase II is evaluated as a communication-link experiment. Consequently, the primary result is the SNR-dependent receiver behavior and the measured relationship between modulation selection, bit recovery, packet success, and throughput.

7.2 Overall Communication Performance

The adaptive receiver has two operating regions. From -4 to approximately 8 dB, noise and fading make the four constellations difficult to distinguish. From 10 dB onward, the modulation decision becomes reliable and the BER and SER decrease sharply. This transition is the most important Phase-II result because it exposes the operating limit of the complete communication chain.

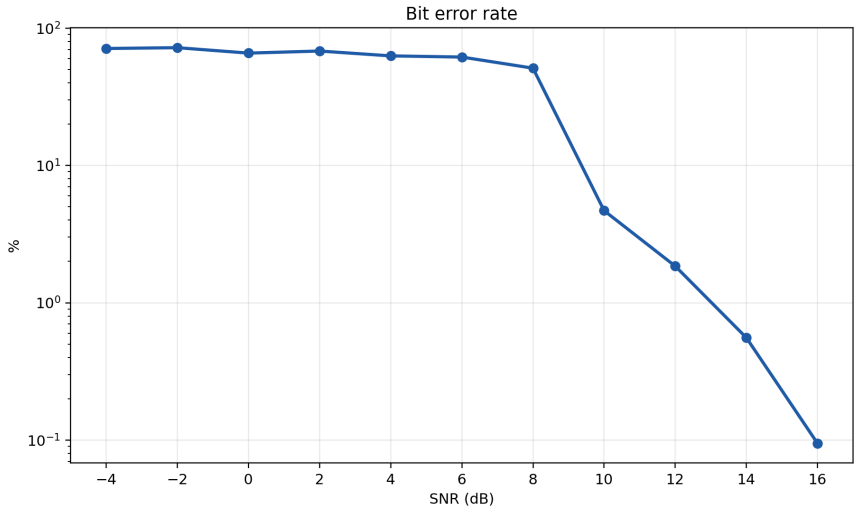


Figure 7.1: Adaptive receiver BER over the complete SNR sweep.

7.3 SNR-Wise Performance

At 10 dB, the decision accuracy is already 99.5%, but BER remains 4.69%. This demonstrates that correct modulation identification does not guarantee correct symbol decisions. At 12 dB, BER decreases to 1.85% and packet success increases

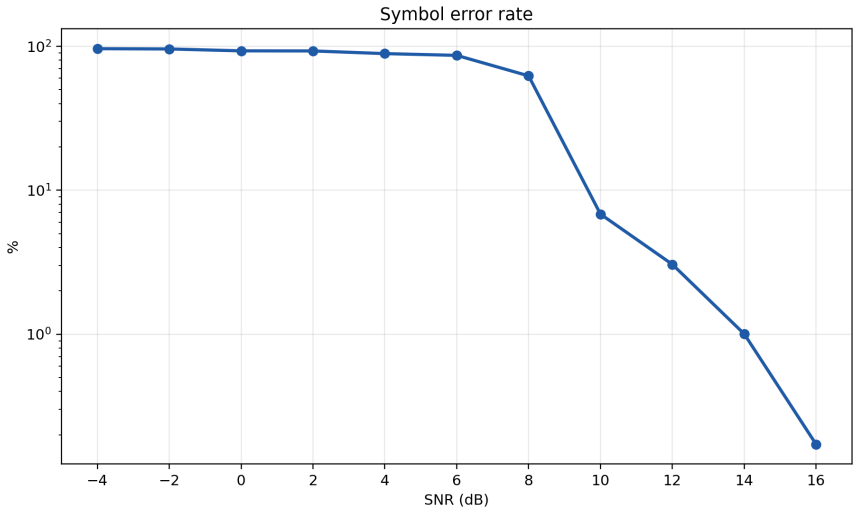


Figure 7.2: Adaptive receiver SER over the complete SNR sweep.

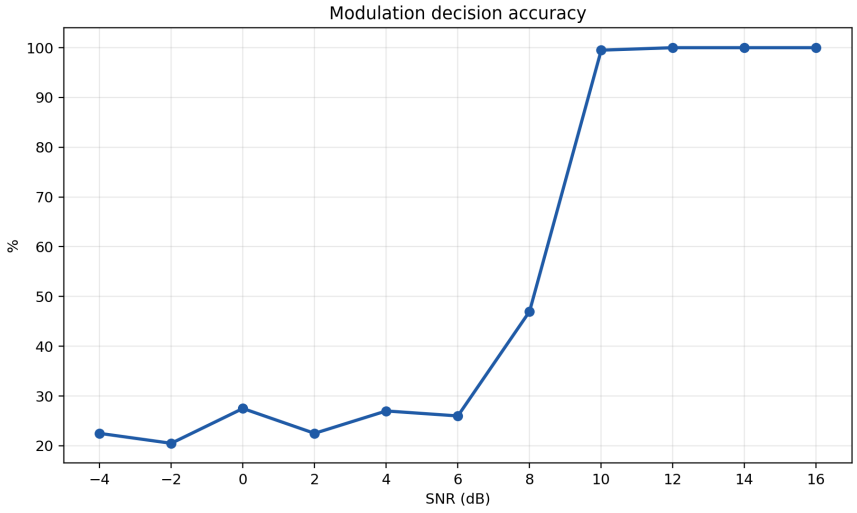


Figure 7.3: Modulation-decision accuracy over the complete SNR sweep.

Table 7.1: Selected SNR-wise results from the final Phase-II experiment.

SNR	**BER**	**SER**	**Decision**	**Packet**	**Throughput**
10 dB	4.69%	6.94%	99.5%	49.0%	1.225
12 dB	1.85%	3.05%	100.0%	54.5%	1.363
14 dB	0.56%	1.00%	100.0%	57.0%	1.425
16 dB	0.095%	0.172%	100.0%	75.5%	1.888

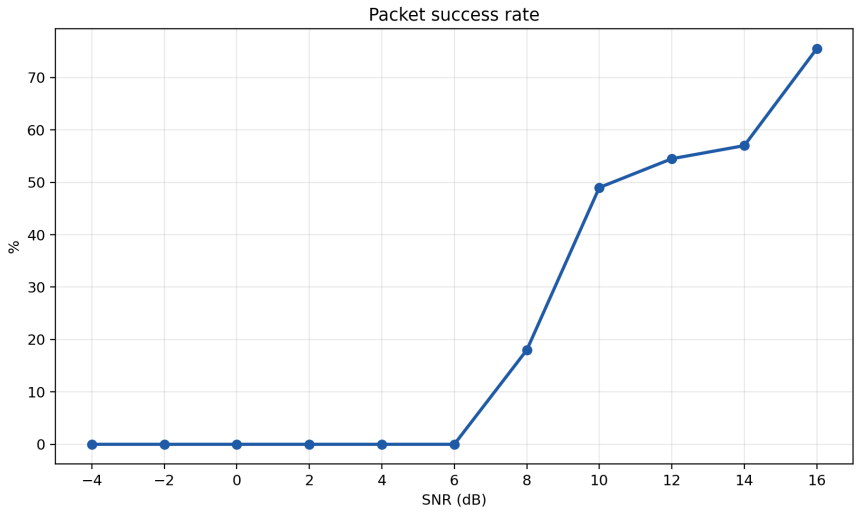


Figure 7.4: Packet-success rate over the complete SNR sweep.

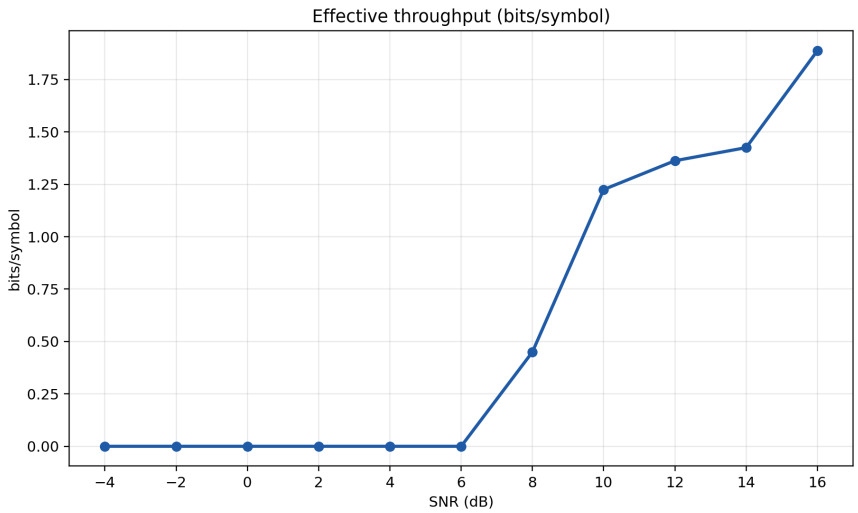


Figure 7.5: Effective Phase-II throughput over the complete SNR sweep.

to 54.5%. At 16 dB, the adaptive receiver enters its most reliable measured operating point.

7.4 High-SNR Performance

At 16 dB, measured BER is 0.095% and SER is 0.172%, with 100% modulation-decision accuracy. Packet success is 75.5% because a packet fails when any bit in its 256-symbol frame is incorrect. Effective throughput reaches 1.888 bits/symbol because successful frames contribute according to their modulation order.

7.5 Complete SNR Sweep

Table 7.2 lists every SNR value in the final sweep. The low-SNR rows show that a receiver can produce a modulation decision for some frames even when no complete packet is recovered. From 8 dB onward the curves begin to separate from the fixed-QPSK reference, and the strongest improvement occurs between 8 and 10 dB. The table also makes clear that the reported 16 dB point is the final measured point of the sweep rather than an isolated best-case trial.

Table 7.2: Complete SNR-wise results for the adaptive Phase-II receiver.

SNR	BER	SER	Decision	Packet	Throughput
−4 dB	71.03%	96.10%	22.5%	0.0%	0.000
−2 dB	72.00%	95.67%	20.5%	0.0%	0.000
0 dB	65.74%	92.81%	27.5%	0.0%	0.000
2 dB	68.07%	92.68%	22.5%	0.0%	0.000
4 dB	62.75%	88.79%	27.0%	0.0%	0.000
6 dB	61.52%	86.35%	26.0%	0.0%	0.000
8 dB	51.07%	62.23%	47.0%	18.0%	0.450
10 dB	4.69%	6.82%	99.5%	49.0%	1.225
12 dB	1.85%	3.05%	100.0%	54.5%	1.363
14 dB	0.56%	1.00%	100.0%	57.0%	1.425
16 dB	0.095%	0.172%	100.0%	75.5%	1.888

7.6 Low-SNR Performance

At −4 dB, adaptive BER is approximately 0.710 and modulation-decision accuracy is 22.5%. Through 8 dB, the estimator remains uncertain and packet success

is zero or low. This region is valuable for analysis because it reveals where an adaptive receiver needs a confidence threshold, fallback to a robust modulation, error-control coding, or retransmission.

7.7 Comparison with Fixed QPSK

The fixed-QPSK receiver maintains BER between approximately 0.57 and 0.65 over much of the sweep because it is mismatched to BPSK, 8-PSK, and 16-QAM frames. The adaptive receiver improves rapidly after the modulation decision transition and reaches 0.000948 BER at 16 dB. This comparison shows that the Phase-II contribution is receiver adaptation, not merely a change in how a label is reported.

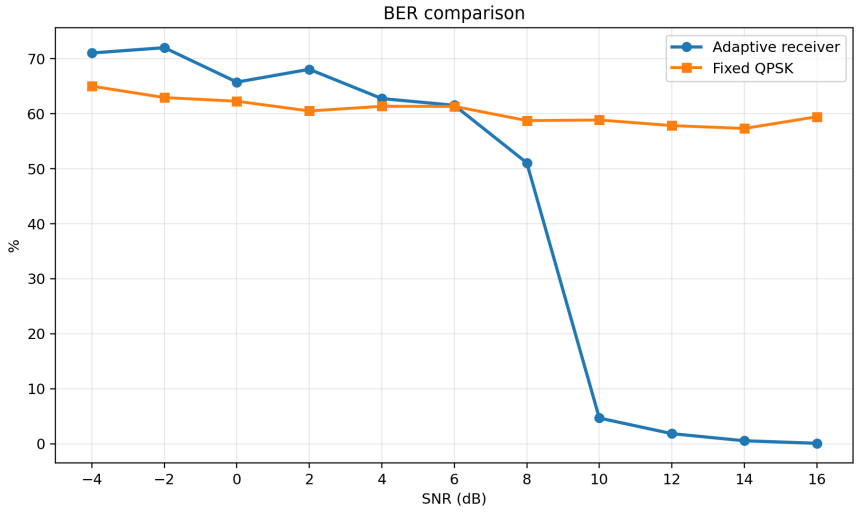


Figure 7.6: Adaptive and fixed-QPSK BER comparison.

7.8 Interpretation by Modulation Order

BPSK contributes one information bit per symbol and has the largest decision margin among the four formats. QPSK increases the information rate while retaining phase-only detection. 8-PSK and 16-QAM carry more information per symbol but are more sensitive to residual phase, amplitude, and noise errors. Since the experiment selects formats across the frame set, the aggregate throughput rises when packet success improves at high SNR.

The fixed-QPSK comparison is intentionally conservative. It is not expected to be optimal for BPSK, 8-PSK, or 16-QAM because it applies the same mapping to every frame. Its relatively flat BER curve therefore indicates the cost of a receiver

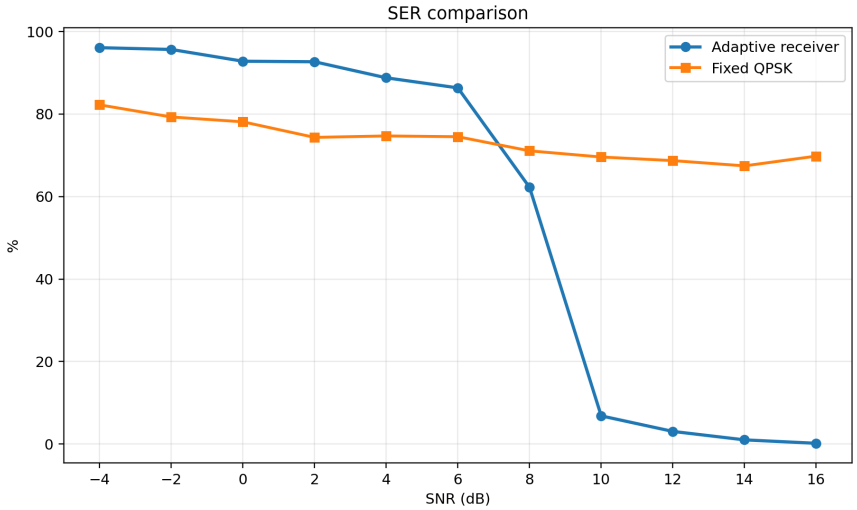


Figure 7.7: Adaptive and fixed-QPSK SER comparison.

that does not adapt, while the adaptive curve shows the benefit obtained after the modulation decision becomes reliable.

7.9 Result Traceability

The individual result figures, complete SNR table, and summary values in the abstract are derived from the same JSON record. The result manifest records the seed, number of frames, modulation set, and best measured operating point. This traceability is important when rounded percentages are copied into tables. The underlying values remain available for rechecking without changing the report text.

7.10 Error Analysis and Observations

Three error sources dominate the results. First, noise causes constellation overlap and estimator ambiguity. Second, fading changes the received amplitude and effective SNR from frame to frame. Third, residual correction error moves points across decision boundaries even when the modulation decision is correct. Packet success combines all of these effects and is therefore stricter than BER or SER.

7.11 Practical Considerations

The current system is suitable for algorithm verification and controlled comparison. A real receiver must estimate channel and carrier parameters from noisy

pilots, handle timing uncertainty, and process samples under latency and memory constraints. The estimator can later be replaced by a trained CNN, but the communication metrics should remain because they measure the usefulness of the recognition decision.

7.12 Limitations of the Results

The experiment uses pilot-assisted simulation correction, flat block fading, an uncoded link, and a constellation-distance estimator. It does not include pulse shaping, timing recovery, frequency-selective multipath, Doppler, oscillator phase noise, nonlinear amplification, or over-the-air hardware. The numerical results should therefore be interpreted as a reproducible Phase-II software baseline.

7.13 Statistical Reliability of the Sweep

Each SNR point contains 200 frames, so the packet-success percentage changes in increments of 0.5 percentage points. This resolution is sufficient to show the transition from no successful packets to a stable high-SNR region, but it should not be interpreted as a confidence interval. A larger run would reduce the variation of the estimate, especially in the low-error region where only a few frames fail.

The BER and SER values are calculated from all bits and symbols in the frames at one SNR, while decision accuracy and packet success are calculated from frame counts. These metrics therefore have different statistical granularities. Reporting them together avoids treating a frame-level percentage as if it had the same uncertainty as a bit-level percentage.

7.14 Engineering Meaning of the Transition Region

The interval from 8 to 10 dB is the most informative part of the measured curve. At 8 dB, decision accuracy is only 47 percent and packet success is 18 percent. At 10 dB, decision accuracy rises to 99.5 percent and packet success rises to 49 percent. This sharp change indicates that the constellation estimator is operating near a separability boundary. A practical receiver could use this region to trigger a conservative fallback, request retransmission, or reduce the modulation order.

7.15 Relationship Between the Individual Curves

The individual plots show related but different aspects of the same receiver. BER and SER decrease when corrected samples remain inside their intended decision regions. Decision accuracy measures whether the receiver selected the correct constellation family. Packet success is stricter than both error rates because one error is enough to fail a frame. Throughput combines packet success with the information carried by the selected modulation.

The curves therefore should not be expected to cross zero at the same SNR. Decision accuracy reaches 100 percent at 12 dB, but BER and SER remain nonzero because some samples are still close to decision boundaries. Packet success increases more slowly because a frame contains hundreds of symbols. This relationship is visible only when the results are plotted separately.

7.16 Interpretation of the Error Floor

The high-SNR point does not produce zero BER. The remaining errors are consistent with the combination of residual correction error, fading realizations, and higher-order modulation points. The result is not an indication that the estimator failed at 16 dB; the decision accuracy is 100 percent there. Instead, it shows that a correct modulation family is necessary but not sufficient for error-free symbol decisions.

7.17 Packet Success and Frame Length

Packet success depends on the number of bits in a frame. A QPSK frame with 256 symbols carries 512 bits, while a 16-QAM frame carries 1024 bits. Consequently, a small average bit error rate can still lead to many unsuccessful frames when the information payload is long. This is why the throughput result is lower than the nominal average bits per symbol and why packet success is retained as a separate engineering metric.

7.18 Usefulness of the Separate Result Figures

The BER, SER, decision-accuracy, packet-success, throughput, and adaptive-versus-fixed comparison figures are generated separately from the same result record. This presentation avoids compressing unrelated scales into one small

image. It also makes each result legible when printed in the report or viewed independently during evaluation.

Chapter 8
Conclusion and Future Work

8.1 Conclusion

This project developed a complete Phase-II adaptive digital communication receiver. The system generates information bits, supports four modulation formats, models channel and carrier impairments, corrects the received sequence, estimates the modulation, selects a demodulator, and evaluates recovered information through communication metrics.

The final experiment shows a clear SNR-dependent operating region. At 16 dB, the adaptive receiver achieves 0.095% BER, 0.172% SER, 100% decision accuracy, 75.5% packet success, and 1.888 bits/symbol throughput. The difference between 99.5% decision accuracy and 4.69% BER at 10 dB also demonstrates why end-to-end communication evaluation is necessary.

8.2 Contributions

- Designed a transmitter–channel–receiver Phase-II architecture.
- Implemented BPSK, QPSK, 8-PSK, and 16-QAM symbol processing.
- Added block-Rayleigh fading, AWGN, phase offset, and frequency offset.
- Implemented correction, constellation estimation, and adaptive demodulation.
- Added a fixed-QPSK reference for system-level comparison.
- Evaluated BER, SER, decision accuracy, packet success, and throughput.
- Created the result record, figures, tests, and supporting project files.

8.3 Limitations

The present receiver is a controlled simulation with pilot-assisted correction and no error-control coding. It does not yet include a complete timing loop, online CNN inference, multipath, Doppler, or SDR measurements. These limitations define the boundary of the current claims rather than invalidating the measured software comparison.

8.4 Future Work

The next stage should implement noisy-pilot timing and carrier recovery, replace the controlled correction with a practical channel estimator, and connect the learned classifier to the live receiver path. Confidence-aware fallback, coding, interleaving, retransmission, frequency-selective channels, Doppler, oscillator phase noise, and nonlinear RF effects should then be evaluated. Finally, the complete system should be tested on an SDR using recorded and over-the-air signals.

8.5 Lessons from the Phase-II Evaluation

The first lesson is that recognition and communication performance should not be treated as identical. The second is that packet success is a useful strict measure for a project that intends to describe a receiver rather than only a classifier. The third is that a controlled simulation can expose the interaction between modulation order, channel quality, and receiver adaptation before hardware resources are introduced.

The evaluation also identified the importance of keeping the experiment record separate from the report narrative. The report uses rounded values for readability, while the result file preserves the full arrays. This separation makes it possible to revise a figure or table without changing the receiver implementation.

8.6 Expected Hardware Demonstration

A hardware demonstration can use an SDR transmitter and receiver with a known pilot sequence and a controlled modulation schedule. The first demonstration should use a cable or attenuated connection to avoid uncontrolled interference. The receiver can then compare the estimated modulation with the configured transmitter mode and calculate BER using a known payload. After the cable test, an over-the-air test can introduce antenna and multipath effects.

The simulation modules already provide the logical stages required for this extension. The channel function would be replaced by sample acquisition, the pilot-assisted correction would become a measured synchronization loop, and the same modulation estimator and metric functions could be retained. This staged transition reduces the risk of changing the algorithm and the measurement method at the same time.

Appendix A
Modulation Class Descriptions

A.1 Summary of Modulation Characteristics

Table A.1 summarizes the four formats used in the Phase-II receiver. The formats are normalized before transmission so that comparison across SNR values is based on the channel condition rather than an arbitrary signal-power difference.

Table A.1: Modulation classes used by the Phase-II communication link.

Class	**Bits/symbol**	**Description**
BPSK	1	Two opposite phase states; robust and low-rate.
QPSK	2	Four phase states; balanced robustness and rate.
8-PSK	3	Eight phase states; higher rate and smaller phase separation.
16-QAM	4	Amplitude and phase states; high rate and higher sensitivity.

A.2 Communication-Metric Definitions

BER is the fraction of recovered bits that differ from the transmitted bits. SER is the fraction of recovered symbols that differ from the transmitted symbols. Decision accuracy is the fraction of frames for which the estimated modulation is correct. Packet success requires every bit in a frame to be correct, while throughput is the successful information rate per transmitted symbol.

Table A.2: Metrics used to evaluate the Phase-II receiver.

Metric	**Meaning in this project**
BER	Fraction of recovered bits that differ from transmitted bits.
SER	Fraction of recovered symbols that differ from transmitted symbols.
Decision accuracy	Fraction of frames with the correct modulation estimate.
Packet success	Fraction of frames with every bit recovered correctly.
Throughput	Successful information bits per transmitted symbol.

Appendix B
Project Workflow Summary

B.1 Phase-II Workflow

The Phase-II workflow consists of constellation construction, information-bit generation, channel modelling, receiver correction, modulation estimation, adaptive demodulation, metric calculation, and result plotting. The source files supplied with the report follow this same order. The workflow begins with a known transmitted frame and ends with a comparison between transmitted and recovered bits.

1. Generate a binary information frame.
2. Select BPSK, QPSK, 8-PSK, or 16-QAM and map the bits to symbols.
3. Apply block-Rayleigh fading, carrier offsets, and AWGN.
4. Correct the simulated channel and carrier effects.
5. Estimate the modulation from the corrected constellation.
6. Select the corresponding demodulator and recover the bits.
7. Calculate BER, SER, decision accuracy, packet success, and throughput.
8. Store the numerical result and generate the individual Phase-II plots.

B.2 Experiment Record

The final experiment uses a seed of 42, 256 symbols per frame, 200 frames at each SNR, and SNR values from -4 to 16 dB in 2 dB steps. The JSON result record stores the complete adaptive and fixed-QPSK arrays used by the tables and figures. It also stores the modulation list and frame count so that the reported curves can be traced to one defined experiment.

B.3 Files Supplied with the Report

The package contains the report source, SVNIT style file, logo, figures, JSON result record, communication module, experiment driver, and tests. The main result figures are stored individually rather than as one combined panel. This allows each graph to be read clearly in the report and used separately in a presentation or review.

B.4 Recommended Operating Regions

The measured curves suggest three practical operating regions. Below approximately 8 dB, the receiver is not dependable for uncoded packet delivery because decision accuracy and packet success are low. Between 8 and 12 dB, the receiver is in a transition region where a confidence threshold, retransmission, or robust fallback would be useful. Above 12 dB, the modulation decision is stable, although residual bit errors still prevent every frame from being correct.

These regions are not universal channel limits. They belong to the selected frame length, modulation set, channel model, correction method, and number of experiment frames. Their value is that they translate the curves into a receiver design decision: the same estimator can be used continuously, but its output should be trusted differently depending on the measured signal quality.

B.5 What the Current Result Demonstrates

The result demonstrates that modulation recognition can be connected to an adaptive demodulator and evaluated with physical-layer metrics. It also demonstrates that the communication result contains more information than the decision label. At 10 dB the label is correct for 99.5 percent of the frames, yet the bit and packet results still show substantial loss. At 16 dB, the label is correct for every frame in the experiment, but the remaining BER confirms that symbol detection and packet delivery must still be measured separately.

B.6 Implications for System Design

If packet delivery is more important than spectral efficiency, the receiver can use the decision margin to avoid high-order formats in the transition region. If throughput is the main objective, higher-order formats can be allowed when the channel estimate and modulation decision are reliable. Forward-error correction and interleaving would change the relationship between BER and packet success, but they would not remove the need for modulation estimation. The present uncoded result is therefore a useful reference before coding is introduced.

B.7 Checks on the Main Numerical Claims

The main numerical claims can be checked directly from the final SNR table. At 16 dB, the BER and SER are below one percent, decision accuracy is 100 percent,

packet success is 75.5 percent, and throughput is 1.888 bits/symbol. At 10 dB, the decision accuracy is 99.5 percent while BER is 4.69 percent. At 8 dB, decision accuracy falls to 47 percent and packet success is 18 percent. These three points cover the high, transition, and low-reliability regions without relying on a single curve.

The fixed-QPSK comparison gives a second check on the adaptive result. At 16 dB the fixed receiver still has 59.45 percent BER, while the adaptive receiver has 0.095 percent BER. Both paths process the same corrected frames, so the difference is produced by demodulator selection rather than by a different channel sweep.

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